

Use the Taylor expansion to find the linear approximation $L(x)$ to $F(x)$ near x_0 , where

$$F : \mathbb{R}^3 \longrightarrow \mathbb{R}^3, \quad \begin{pmatrix} u \\ v \\ w \end{pmatrix} \longmapsto \begin{pmatrix} u + e^{u+w} \\ v + w^2 \\ uv + w \end{pmatrix}, \quad x_0 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}.$$

The linear approximation of F at x_0 is

$$L(x) = F(x_0) + DF(x_0)(x - x_0).$$

First,

$$F(x_0) = F \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} e^{-1} \\ 2 \\ -1 \end{pmatrix}.$$

The Jacobian matrix of F is

$$DF(u, v, w) = \begin{pmatrix} 1 + e^{u+w} & 0 & e^{u+w} \\ 0 & 1 & 2w \\ v & u & 1 \end{pmatrix}.$$

Therefore,

$$DF(x_0) = \begin{pmatrix} 1 + e^{-1} & 0 & e^{-1} \\ 0 & 1 & -2 \\ 1 & 0 & 1 \end{pmatrix}.$$

Since

$$x - x_0 = \begin{pmatrix} u \\ v - 1 \\ w + 1 \end{pmatrix},$$

the linear approximation is

$$L(u, v, w) = \begin{pmatrix} e^{-1} \\ 2 \\ -1 \end{pmatrix} + \begin{pmatrix} 1 + e^{-1} & 0 & e^{-1} \\ 0 & 1 & -2 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} u \\ v - 1 \\ w + 1 \end{pmatrix}.$$

Equivalently, after simplifying,

$$L(u, v, w) = \begin{pmatrix} (1 + e^{-1})u + e^{-1}w + 2e^{-1} \\ v - 2w - 1 \\ u + w \end{pmatrix}.$$